"Credit scores are treated as objective measures of risk, yet one recent NBER study finds they misclassify **41% of consumers**, with errors hitting the most vulnerable hardest: **47% of Subprime and 70% of Near Prime borrowers** are placed in the wrong risk category."

SOURCE: Albanesi S. & Vamossy D. F. (2024), NBER WORKING PAPER SERIES

# NovaCred

MODEL GOVERNANCE AND DATA AUDIT

GROUP 3

# The (Dream)Team

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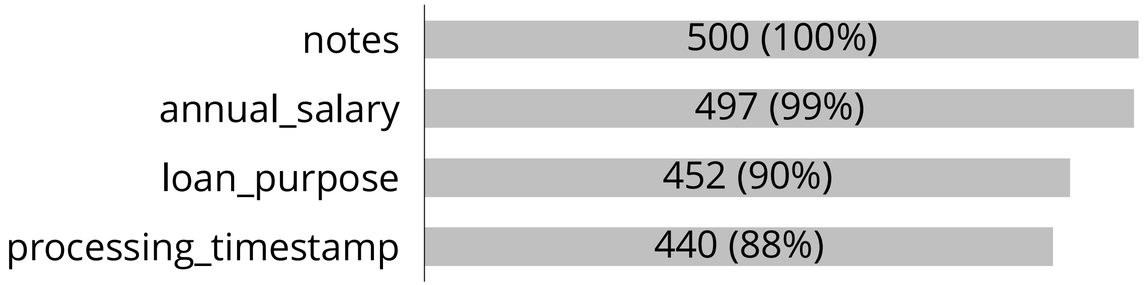
# Missing values for decision-critical fields and structural inconsistencies pose moderate risk to Novacred

## Completeness

Moderate

**Top 10 missing fields (w/o "spending_" fields)**



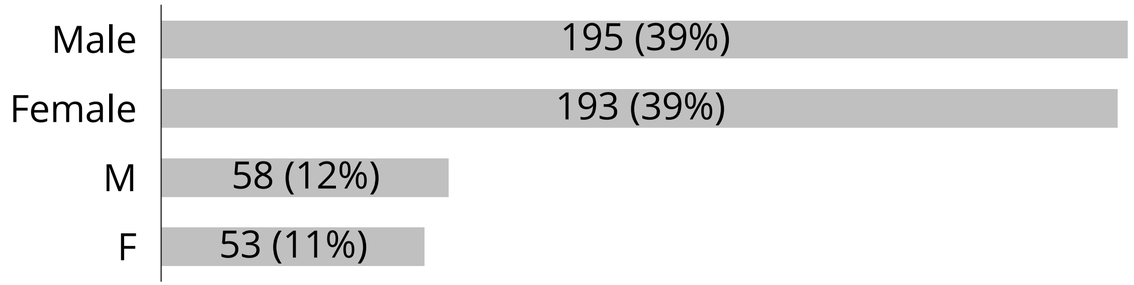

**Key Findings**

- **>80% of fields have missing values** (>50% "spending_" fields)
- **Auditability and business context** is the main gap: processing timestamp 87.6% and loan purpose 90% missing
- **Schema inconsistency:** annual_income vs. annual_salary
- **Missingness is understated**: empty strings across 14 values
- **Traceability gaps** : ssn and ip_address missing in 1.0% each

## Consistency

Moderate

**Exp.: Inconsistent gender encoding (3 records NaN)**




**Key Findings**

- **Primary key integrity breach**: 2 duplicated IDs (4 records)
- **Data type inconsistency** annual_income (488 int, 8 str, 1 float)
- **Duplicate-record divergence:** minor within-ID inconsistencies (e.g., SSN missing in one duplicate)
- **Non-standard date formats:** 3 coexisting date_of_birth formats (340 + 101 + 56 records)

Validity

Accuracy

# Invalid values and formats and inaccuracies in decision-critical variables pose moderate-high risk to Novacred

## Completeness

### 1 Invalid values

- **Domain violation in DTI:** 1 record (0.2%) with 1.85
- **Implausible values for decision-critical fields:** 2 records (0.4%) with <0 credit history and 1 (0.2%) record with <0 savings balance[1]

### 2 Invalid formats

- **Invalid contact data:** 1 record (0.2%) with malformed email (missing @)
- **Schema/type mismatches:** Misaligned fields processing timestamp, annual income (object vs. datetime, object vs. float)

Validity — Moderate

## Consistency

### 1 Approved loans w/o or 0 income

- **Decision integrity breach:** 3 approved loans (0.6%) have zero or missing income (due to schema mismatch identified in Completeness)

### 2 Total spending exceeding reported income

- **Income–spending inconsistency:** 1 record (0.2%) reports income = 0 but total spending = 1,389

Accuracy — Moderate - High

# Bias analysis confirms invalid loan approval rates by gender

## Analysis

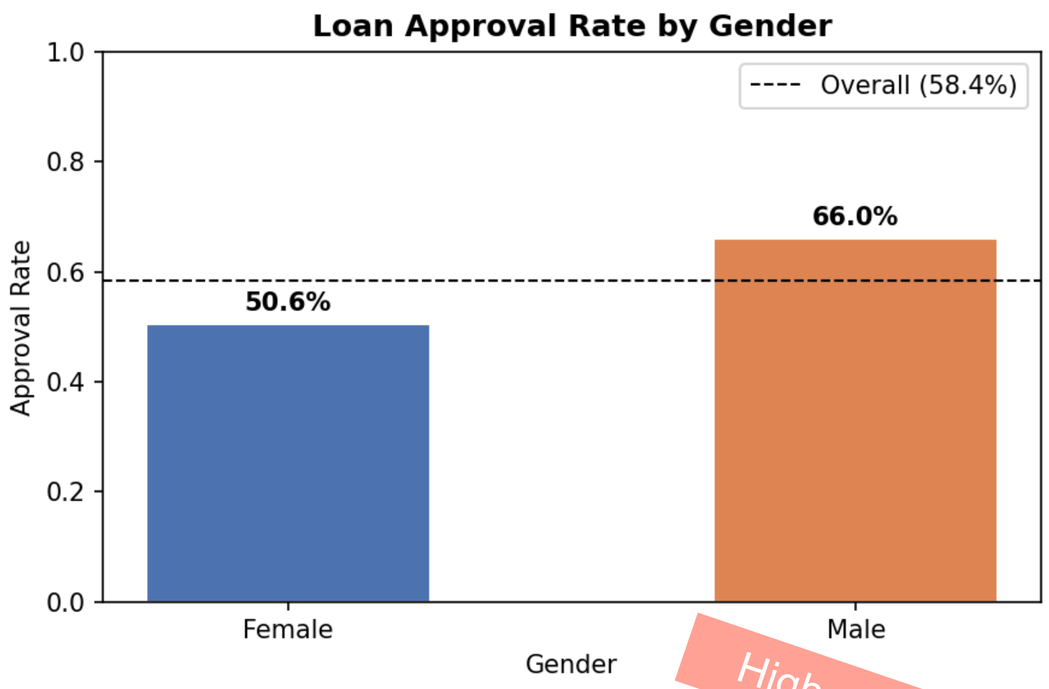
### 1 Baseline Disparity Analysis

- Approval Rate for females lies **15,4 Percent points** below the one for males
- Disparate Impact Ratio is **0.77 below four-fifths** threshold of 0.8

### 2 Conditional Fiarness Analysis

- **Odds ratio** for male is at **1.98** at ceteris paribus of financial numbers

**Regulations: Discriminatory Decisioning under GDPR Article 22 and EU AI Act Annex III.**




High - Critical

# Deeper Analysis confirms further bias risks through proxy variables

## Key Bias Findings with Conclusion

| | | | |
|---|---|---|---|
| **Intersectional Effects (Age x Gender)** | High | Female 26–35: DI = 0.62; Female 18–25: DI = 0.77; Female 51–65: DI = 0.76 | Overall DI of 0.77 understates subgroup harm — all three violate four-fifths rule |
| **ZIP Code Proxy** | High | NYC 88.8% male; LA 93.5% female; $\chi^2 = 324.67$, $p < 0.001$ | Not an active proxy (OR = 1.14, p = 0.67) but encodes gender — GDPR Art. 5(1)(c) breach |
| **Age Discrimination** | Moderate | Approval: 43.5% (18–25) → 67.4% (36–50); KW p = 0.003 | Not independent after controls (OR = 1.00, p = 0.72) — reflects risk profile differences |
| **Sensitivity Spending Data** | Moderate | Gambling, adult entertainment & alcohol categories collected Male | No demonstrated credit-decision necessity — likely violates GDPR Art. 9 |
| **Pricing Fairness** | Low | 4.63% vs. Female 4.49%; Δ = +0.14 pp · OLS p = 0.288 | No statistically significant pricing discrimination detected |

# PII direct and quasi identifiers with almost 100% coverage and (conditional) sensitive categories pose governance risks

## PII Identifiers

| | | | |
|---|---|---|---|
| full_name | 100% | **Direct identifier** | Critical |
| email | 98% | **Direct identifier** | Critical |
| ssn | 99% | **Direct identifier** | Critical |
| ip_address | 99% | **Direct identifier** | Critical |
| date_of_birth[1] | 100% | **Quasi identifier** | High |
| zip_code[1] | 100% | **Quasi identifier** | High |
| gender[1] | 100% | **Protected Attr.** | High |
| sensit. spend. cat. | 5% | **Cond. sensitive** | High |
| loan purp. = medical | 2% | **Cond. sensitive** | High |

## GDPR Mapping

- **Purpose limitation / Data minimisation (Art. 5(1)(b), 5(1)(c))**
  sensitive behavioural fields lack clear necessity
- **Lawfulness / Transparency / Accountability (Art. 6, 13, 5(2))**
  lawful basis and notice traceability are not evidenced
- **Automated decision safeguards (Art. 22)**
  rejection explanations are largely opaque
- **Privacy by design / Confidentiality (Art. 25, 5(1)(f))**
  direct identifiers remain in analytics data
- **Storage limitation / Accountability (Art. 5(1)(e), 5(2))**
  retention and deletion controls are not evidenced

RECOMMENDATIONS

# Prioritizing the most severe issues, we recommend 4 legally required measures for Novacred

## Before deployment (legally required)

**1**

**Suspend automated approvals and remediate gender bias** (DI = 0.77, OR = 1.98 after controls)
No deployment until DI exceeds 0.80

**2**

**Complete DPIA under GDPR Art. 35** with necessity assessment, risk assessment, and documented safeguards.

**3**

**Enforce privacy by default**
separate identity store, replace opaque rejections with contestable reason codes, implement consent mechanism with versioning and withdrawal handling

**4**

**Implement EU AI Act Art. 9-15:**
Incl. risk management system, audit trail with timestamps and decision logging (Art. 12), transparency documentation (Art. 13), human oversight (Art. 14)

## After deployment

- **Ongoing fairness monitoring** with escalation thresholds (DI < 0.85 warning, DI < 0.80 mandatory review).
- **Data retention policy** with automated deletion, deletion audit logs, and DSAR workflow.
- **Governance framework**
  AI system owner, release gates, periodic compliance review.

CONCLUSION

# Novacred cannot deploy before properly introducing governance, quality and bias reduction mechanisms

## 1 Data Quality

**16** issues identified

Overall risk: Moderate - High

## 2 Data Bias

**0.77 = DI**

**1.98 = OR**

Overall risk: High

## 3 Privacy and Governance

**Direct** identifiers unprotected

**6/7** AI Act obligations unmet

Overall risk: Critical